



THOMAS LABETOUILLE

Software & Embedded Systems Developer | Gameplay Programmer

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[LinkedIn](#) • [Portfolio](#) • Driving license — own vehicle

PROFILE

Software developer combining hands-on gameplay programming (Unreal Engine 5, C++) with embedded simulation development for the defence sector (C#, Unity). Recent expertise in integrating AI agents (Claude AI / MCP) into UE5 development workflows. Seeking a permanent or fixed-term position — fully remote or Toulouse-based.

PROFESSIONAL EXPERIENCE

Embedded Systems Technician — CS Group

Sep. 2024 – Nov. 2025

- Designed and integrated interactive scenarios for aeronautical simulators (Inscape VTS / national defence), meeting strict functional and safety requirements.
- Developed modular software architectures, debugged complex simulated behaviours, handled classified documentation.
- Collaborated with subject-matter experts (system engineers, pilots) in a regulated defence environment.

Gameplay Programmer (Freelance) — Rainbow Ant Studio

Dec. 2023 – Sep. 2024

- Built core character gameplay mechanics, swimming system, dynamic weather integration and interactive map in Unreal Engine 5.
- Integrated animations via State Machines & Blend Spaces; wrote progress reports in a fully remote team.

Personal Project — AI Agent × UE5 — Claude AI + Unreal Engine 5

2025–2026

- Bidirectional MCP bridge between Claude Desktop and UE5 via Remote Execution API (C++ / Python): actor spawning, Blueprint creation, NavMesh rebuild — all driven by conversation.
- Native C++ `LampDetectionComponent`: real-time detection of an AI enemy inside a `SpotLight` cone (dot product, line trace) with direct Blackboard writes via `AAIController`.
- 4-state Behavior Tree (Illuminated · Stunned · Chase · Patrol) and procedural horror level built in 3 acts via Python UE5 scripts.

TECHNICAL SKILLS

Engines & languages: UE5, Unity, C++, C#, Python, Blueprint (UE5)

AI & Tools: Claude AI, MCP (Model Context Protocol), Remote Execution API UE5, AI-driven development pipeline

Version control: Git (GitHub, Perforce) • **Cross-cutting:** Team leadership, autonomous remote worker, technical English, classified documentation

EDUCATION & CERTIFICATIONS

Bachelor's Degree in Video Game Development — E-Artsup

2020 – 2023

RNCP Level 6 Certification — Specialisation: C# / Unity / Game & Level Design • Vocational Baccalaureate in Commerce — with distinction

ADDITIONAL INFORMATION

Nationality: French • **Native language:** French • **Technical English:** Reading / Writing

Interests: Video games (Overwatch e-sport team leader), dance, boxing, fitness.